How to set up a duplicate copy of a file vault on a separate system from a database backup?

The following instructions describe in detail how to restore a file vault copy from a SQL database backup (.bak) file on a separate server in order to recreate a working duplicate copy of the original PDMWorks Enterprise vault. This is useful when you for example need to test functionality in a new version or service pack, without disrupting and changing the original production vault. You can also use this method when planning new changes to the production vault in order to verify that everything works as expected before implementing the changes in the live environment.

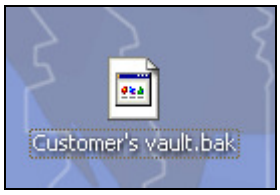
Note that these instructions imply that you set up the file vault on a separate system than where the original vault is set up. It is possible to use the same SQL server as hosting the original vault database, but in that case it is important that you give the restored vault database copy a unique database name. You cannot however have multiple copies of the same file vault on the same archive server – each vault name needs to be unique. Also note that if you host multiple copies of the same source vault on the same SQL server, the index server will not function correctly.

The following steps should be performed:

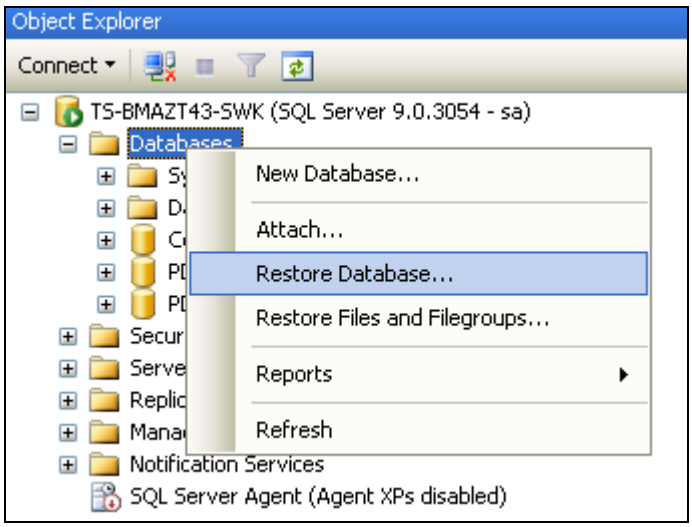
1. Restore the file vault SQL database backup 2
2. Check the file vault database version 5
3. Run an upgrade of the file vault database (if needed): 6
4. Clean up replication information 8
5. Recreate the Linked Server for content searching 9
6. Update the ConisioMasterDb database 10
7. Create the missing Archive Server registry entries and file archive folders 11
8. Overwrite with original file vault archives (optionally) 12
9. Attach to the restored file vault 13
10. Repair user logins 15
11. Remove custom add-ins and dispatch 16

1. Restore the file vault SQL database backup

Ensure you have a complete SQL backup of the original file vault database. *Note that it is possible to work in the restored file vault copy without having the original file vault archives – however you will not be able to use (get) any of the original physical files found in the vault in that case.*



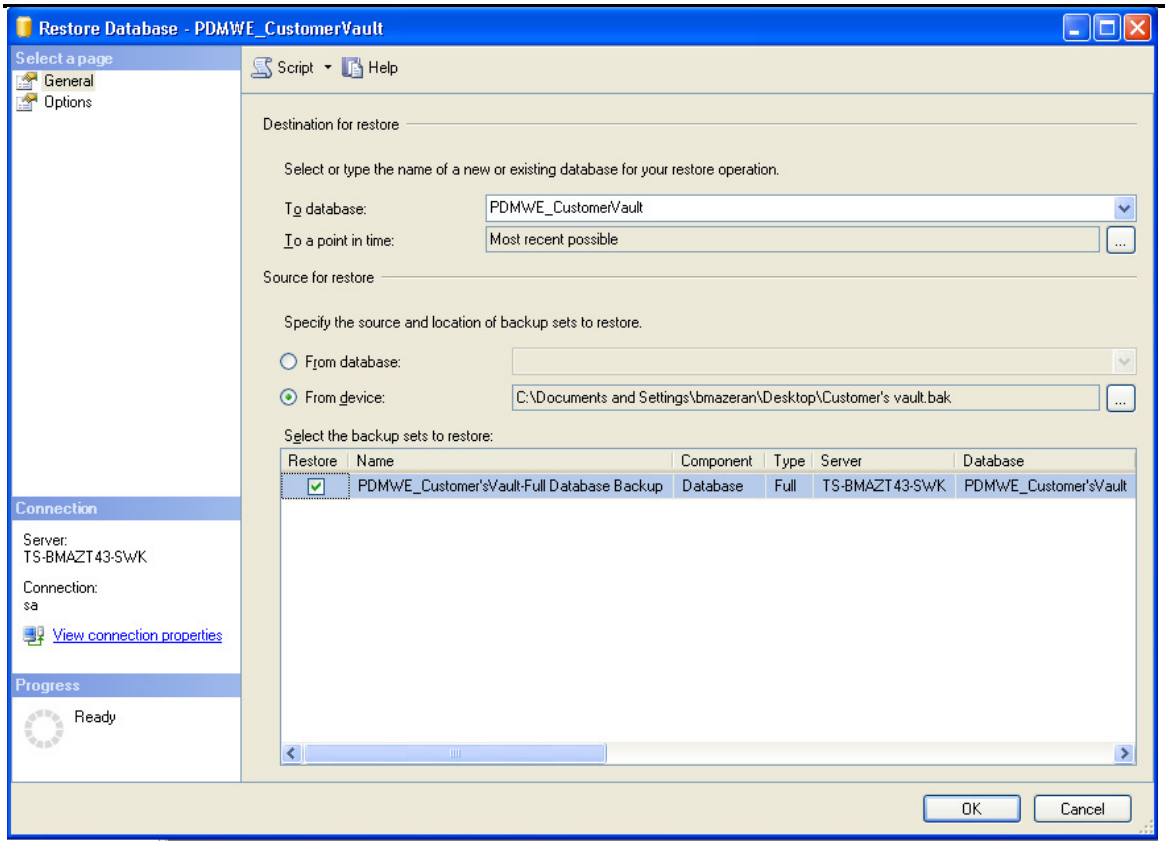
Login as user ***'sa'*** in SQL Server Management Studio, expand the **Databases** node, right-click and pick **Restore Database …**



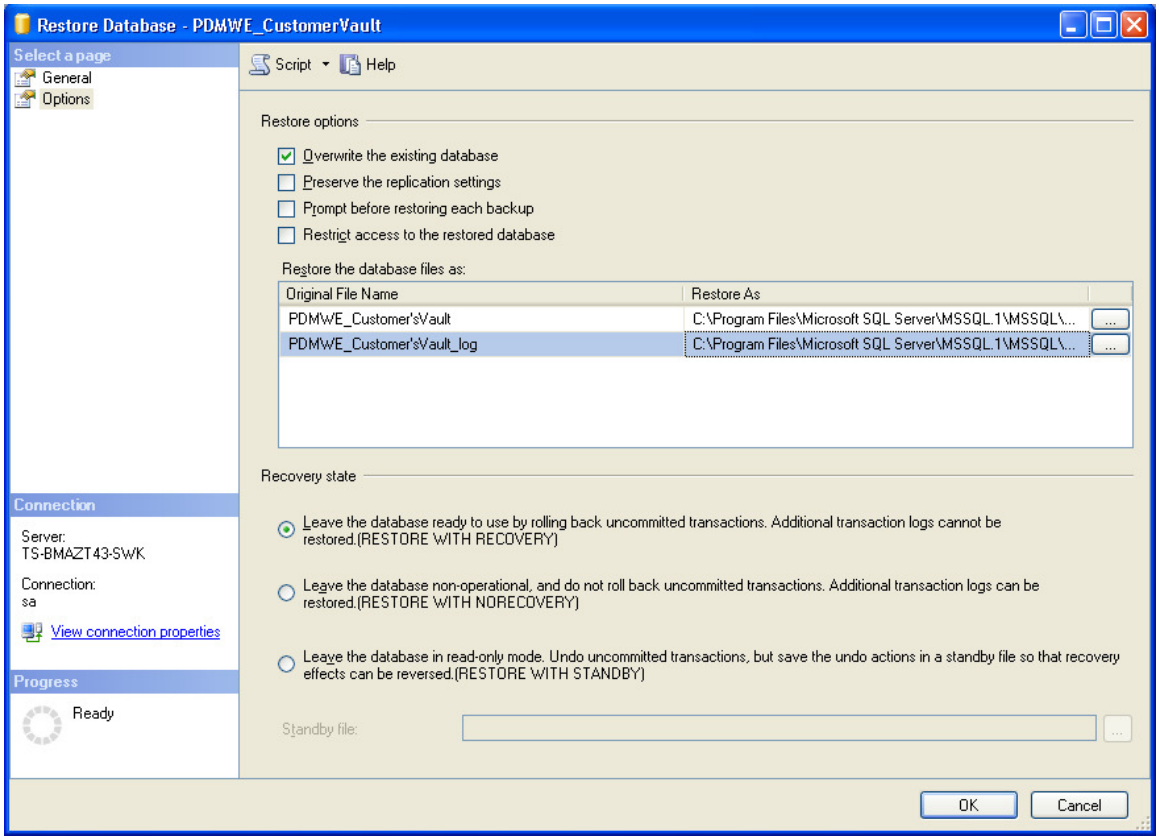
Under **Select or type file name of a new existing database for your restore operation**, select the **To database** drop-list and select the last database name, it should be the name of the exported database (ensure to enter a unique database name if there is already a database with same name as the restored database).

Select **From device**, and navigate to the backup file on your desktop.

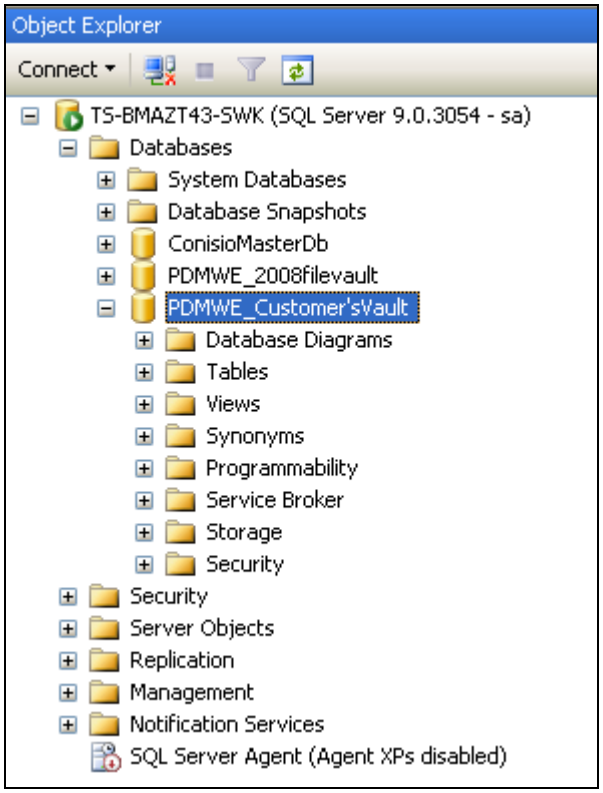
Under **Select the backup sets to restore**, tick the **Restore** checkbox.



Click on the **Options** page, under the **Restore options**, select **overwrite the existing database** (this option should always be picked as a precautionary measure).

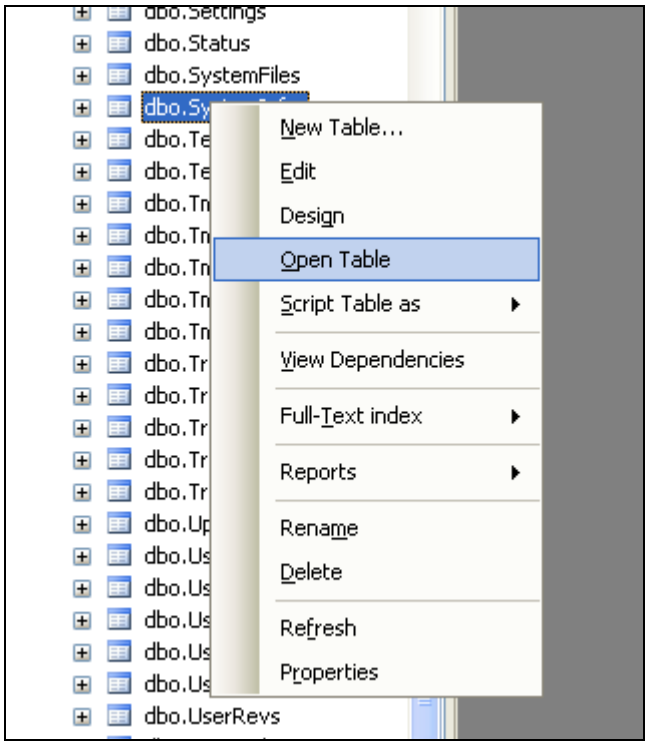


Click **OK** to complete the restore operation. Once complete, refresh the **Databases** node, the customer's file vault database should now be listed.



2. Check the file vault database version

Locate the **SystemInfo** table in the restored database, right-click **Open table**, and take a note of the **Version** column value.



The version number indicates which PDMWorks Enterprise version the database is used with:

PDMWorks Enterprise 2007 SP3	703
PDMWorks Enterprise 2007 SP4	704
…	
PDMWorks Enterprise 2008 SP0	800
PDMWorks Enterprise 2008 SP1	801
…	

…and so on.

In this example, the file vault database backup was from a PDMWorks Enterprise 2007 SP 4 vault i.e. **704**.

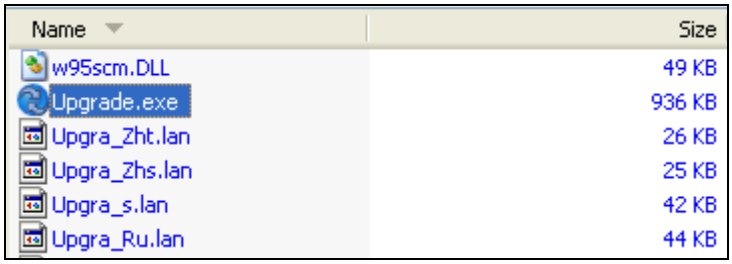
TS-BMAZT43-S...bo.SystemInfo

Version	ArchiveTableVe...	ReactorTableV...	DateFmt	VaultName	ArchiveServerN...	ArchiveServerP...	ArchiveServerV...	RightChangeCo...	LoginLock
704	0	0	120	Customer's Vault	TS-BMAZSRV-SWK	3030	<local>\Archives	1	False
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Run an upgrade of the file vault database (if needed):

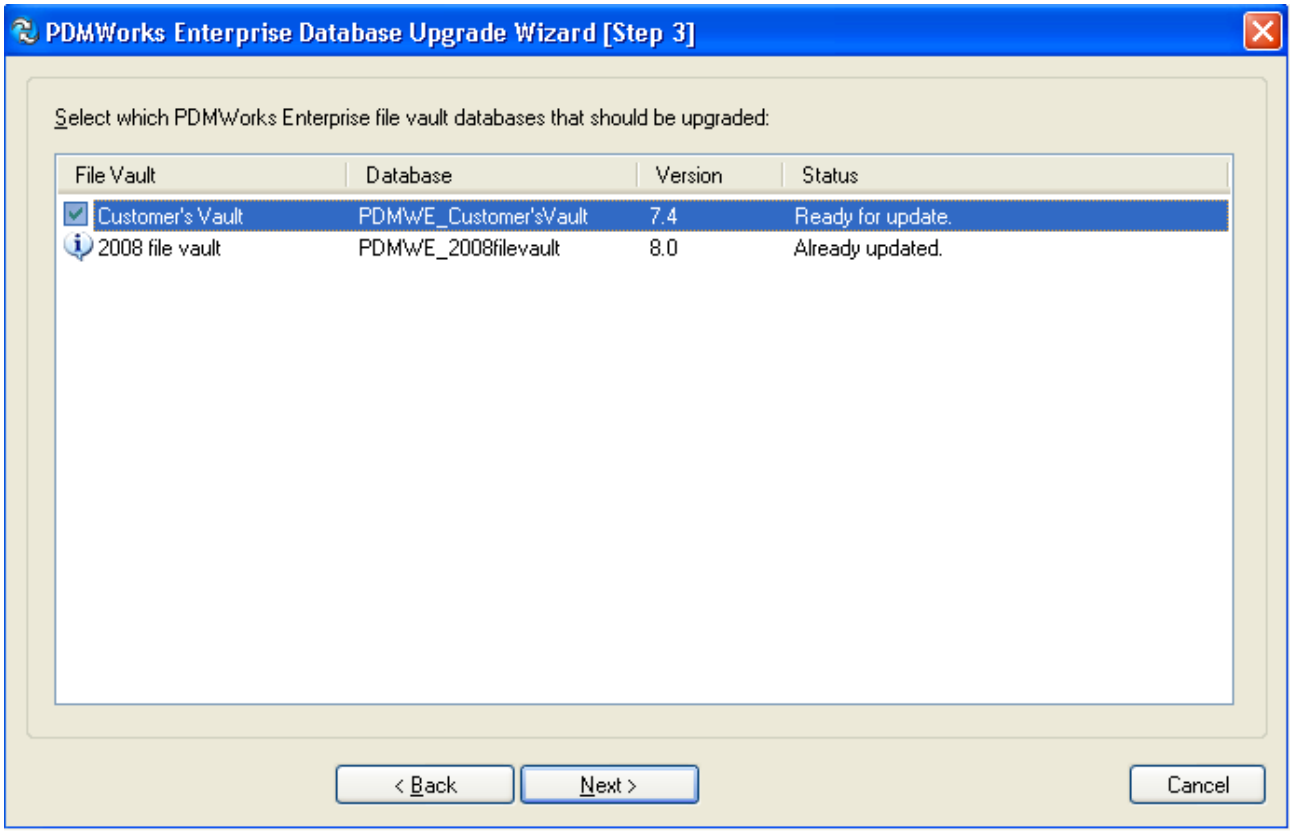
If the vault database is of an older version than currently installed PDMWE version, you need to upgrade the database in order to use the vault. *If not, you need to install same client and server version as the vault copy uses.*

Run **Upgrade.exe** under **Upgrade** folder on the install media.

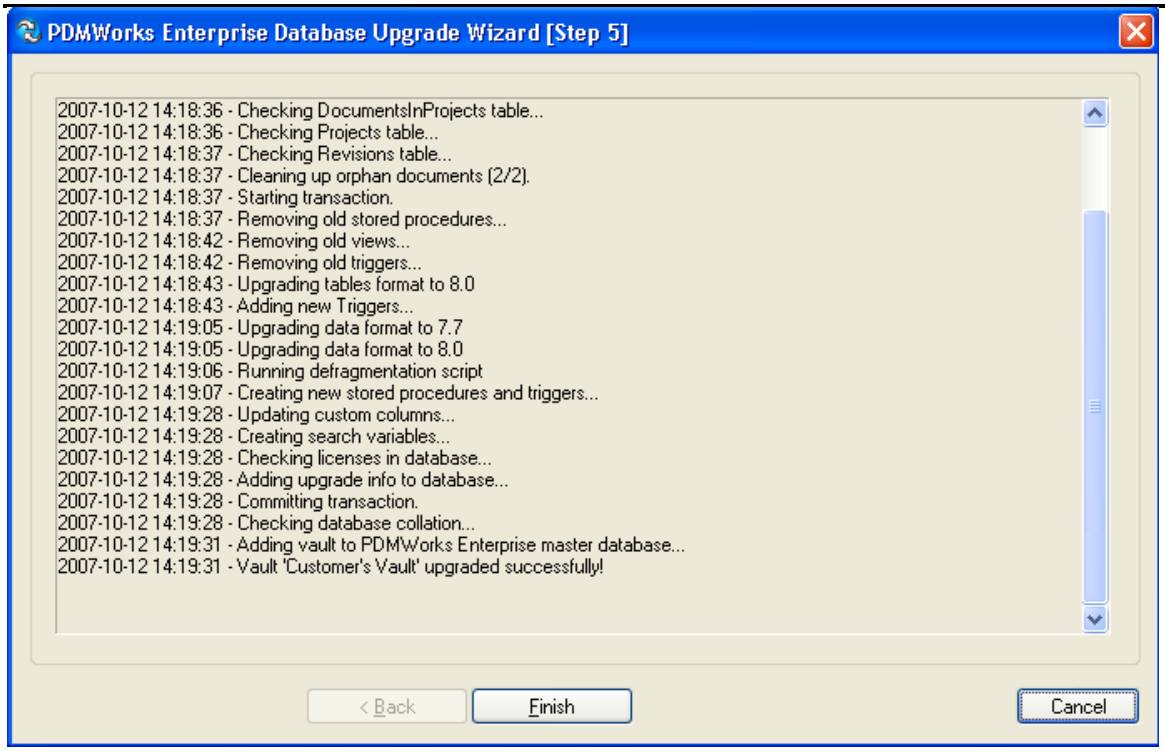


Name	Size
w95scm.DLL	49 KB
Upgrade.exe	936 KB
Upgra_Zht.lan	26 KB
Upgra_Zhs.lan	25 KB
Upgra_s.lan	42 KB
Upgra_Ru.lan	44 KB

Follow the database upgrade wizard to upgrade the vault database to current version.



Once the Upgrade completes successfully, click on **Finish**.



You can verify the upgraded database version by viewing the **SystemInfo** and **Upgrades** table.

Notice that the database version in this example is now 800 (2008 SP0).

TS-BMAZT43-S...bo.SystemInfo

Version	ArchiveTableVe...	ReactorTableV...	DateFmt	VaultName	ArchiveServerN...	ArchiveServerP...	ArchiveServerV...	RightChangeCo...	LoginLock
800	0	0	120	Customer's Vault	TS-BMAZSRV-SWK	3030	<local>\Archives	1	False
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

A new record exists in Upgrades table to list the upgrade from 7.4 to 8.0.

TS-BMAZT43-S... dbo.Upgrades | TS-BMAZT43-S...bo.SystemInfo

UpgradeNumber	Name	Started	Completed	Type	VerMaj	VerMin	Description
1	StoredProcsRegister20071012,14:08:43 (22051828)	10/12/2007 9:0...	10/12/2007 9:0...	1	7	4	Stored procedur...
2	StoredProcsRegister20071012,14:19:05 (2853453)	10/12/2007 9:1...	10/12/2007 9:1...	1	7	4	Stored procedur...
3	StoredProcsRegister20071012,14:19:28 (2876734)	10/12/2007 9:1...	10/12/2007 9:1...	1	8	0	Stored procedur...
4	Upgrade from Ver 7.4 to 8.0	10/12/2007 9:1...	10/12/2007 9:1...	0	8	0	Upgraded by PD...
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

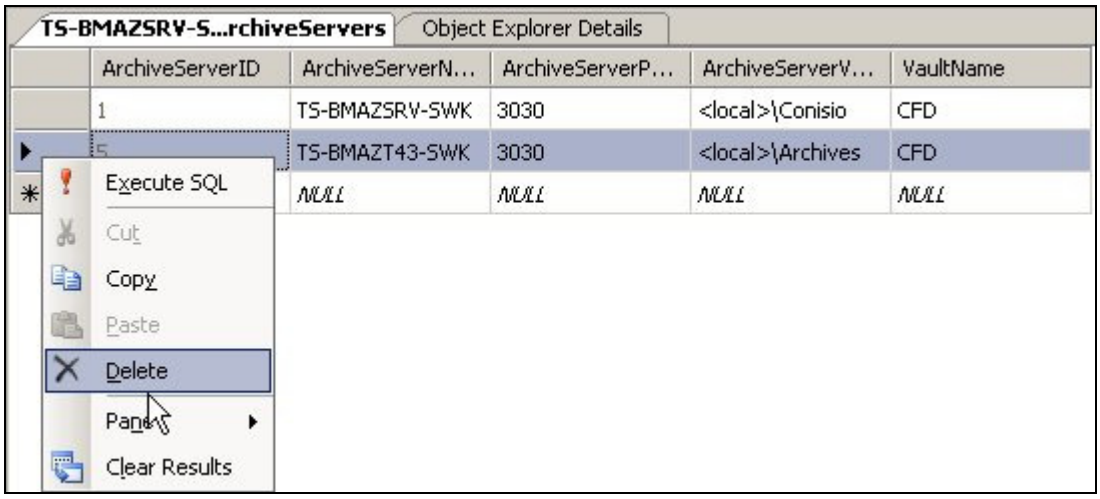
4. Clean up replication information

If the original file vault was replicated, the restored database has entries for the archive servers that are replicating the vault. Those database records will have to be deleted so that the restored vault copy will not try to locate them.

If you wish to test replication in the restored copy it is recommended that you create a new replication once the vault copy is correctly up and running. If you intend to include the whole file vault archive, then all files should be replicated "back" to the main server where the file vault archive copy is taken from before making the database and archive backup. This ensures all archive file versions are available in the restored vault copy.

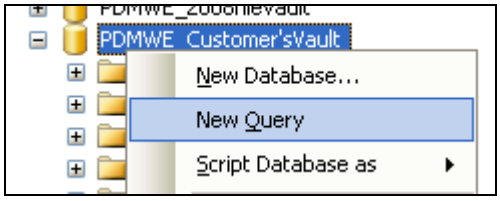
Locate the **ArchiveServers** table in SQL Server Management Studio, RMB, **Open Table**.

Select the replicated archive servers and pick **Delete**. Note that you should not delete the first entry having ArchiveServerID = 1 – this record should show the name of the new archive server you have set up the vault copy on.

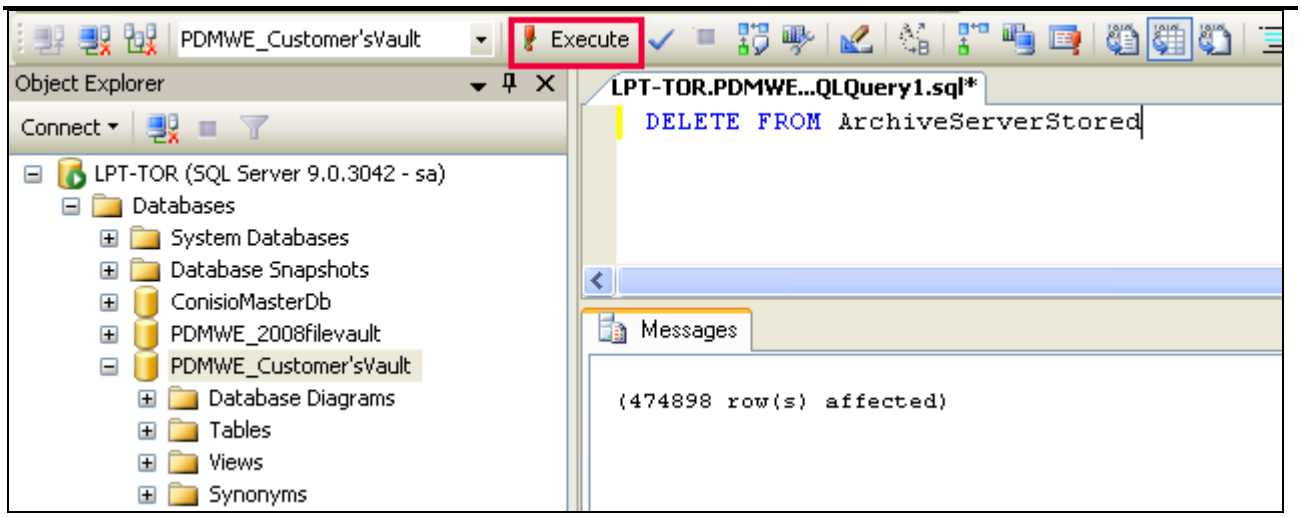


The "**ArchiveServerStored**" (and "**ArchiveServerDeleted**" in PDMWE 2007) tables must be emptied as well to emulate a non-replicated database. i.e.:

In SQL Server Management Studio, right-click on the restored database and select **New Query**



Type the following statement in the query window and press the execute button (or Query, Execute) :
DELETE FROM ArchiveServerStored



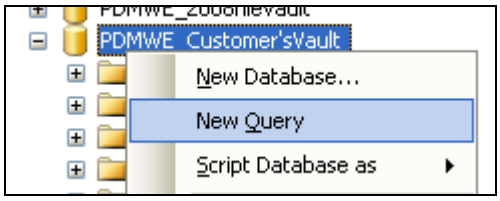
(For PDMWE 2007 or older database you should also run the following query: **DELETE FROM ArchiveServerDeleted**)

5. Recreate the Linked Server for content searching

The SQL server needs to have a linked server set up for the restored vault database in order for indexing / content search to work with the vault.

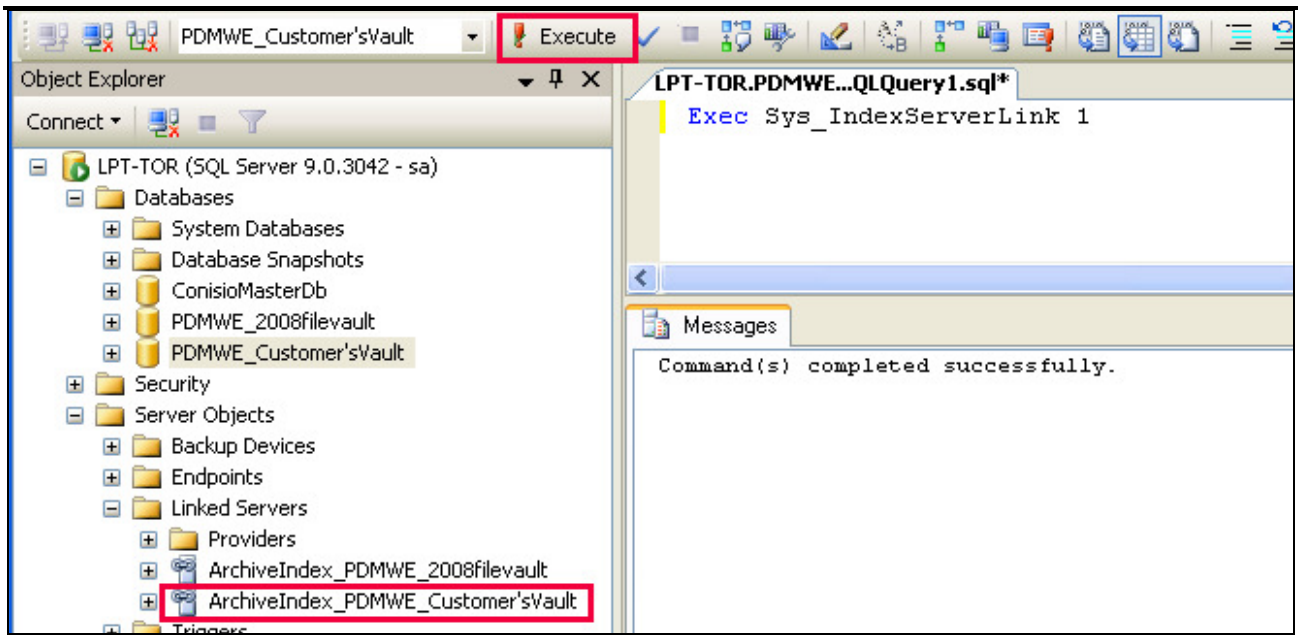
To allow indexing of the file vault database, create a linked server entry:

In **Microsoft SQL Server Management Studio**, right-click on the file vault database and select **New Query**



Enter the following statement in the query window, then press **Execute** button (or Query, Execute) to run the query:

Exec Sys_IndexServerLink 1

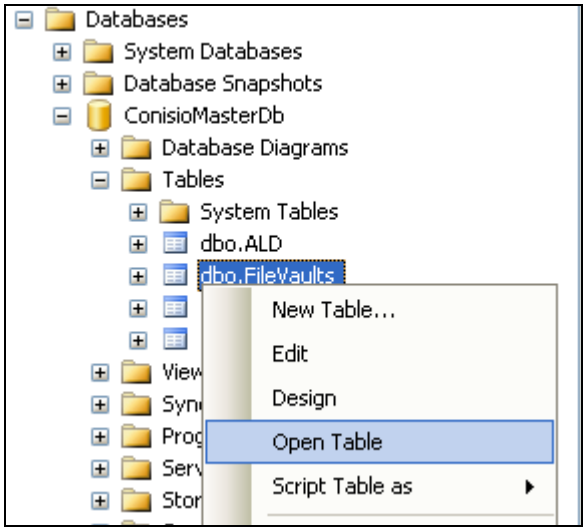


6. Update the ConisioMasterDb database

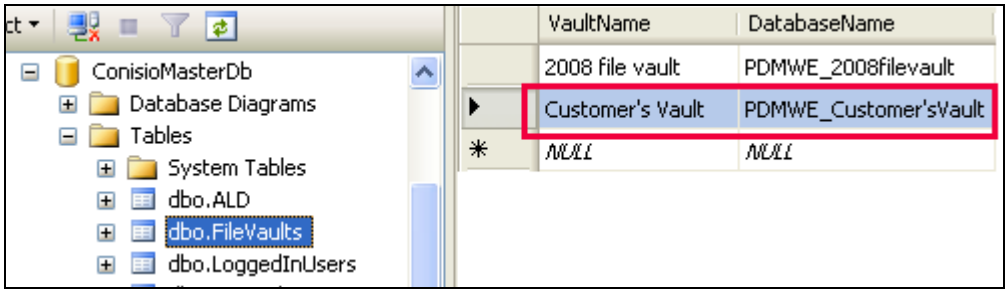
The ConisioMasterDb database contains license information about the file vaults hosted on the SQL server. If you have existing file vaults already on the SQL server you are restoring the vault database copy to, there will already exist a ConisioMasterDb database.

If the SQL server <u>does not</u> yet have a ConisiomasterDb, it is recommended that you also restore a copy of that database from the original server where the vault database was backed up from. Alternatively, you can create a new ConisioMasterDb by using the PDMWorks Enterprise administration tool on a client and creating a temporary new file vault on the SQL server (the initial vault database creation will also create the ConisioMasterDb database).

Open the **FileVaults** table in the **ConisioMasterDb** database.



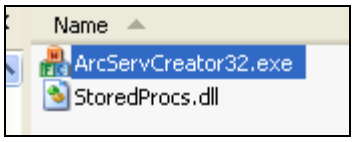
Ensure there is an entry with the restored database name and corresponding file vault name.



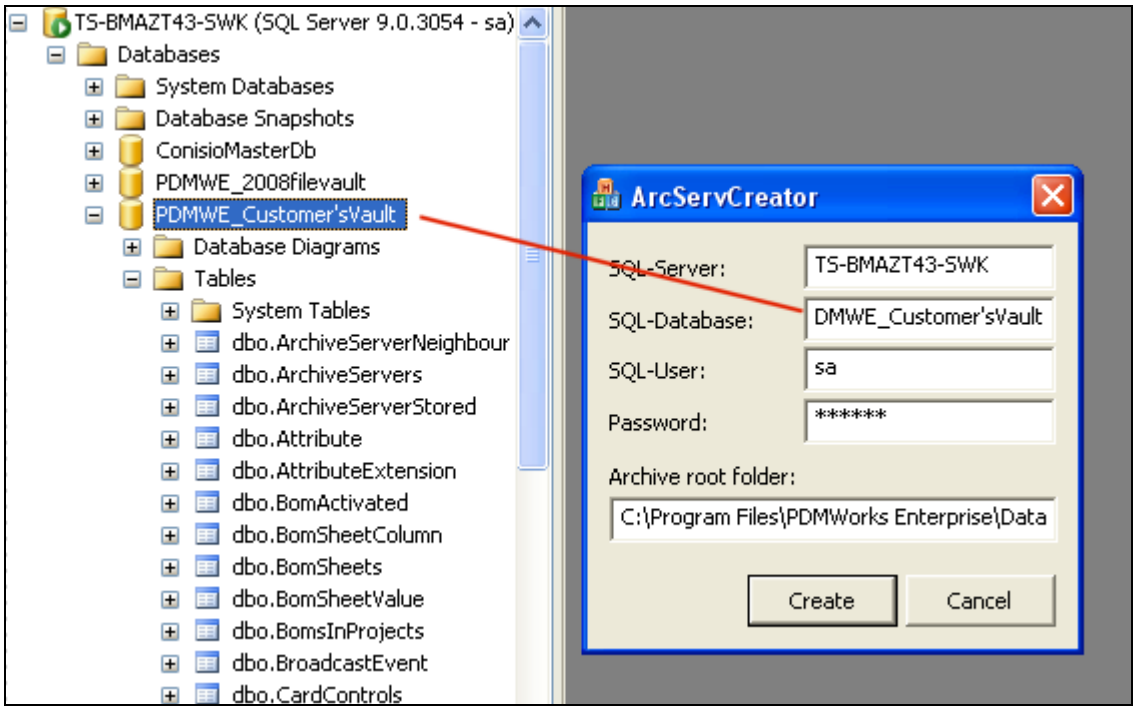
7. Create the missing Archive Server registry entries and file archive folders

The archive server registry needs to contain the vault specific connection information in order to attach and use the restored vault copy – a file vault archive folder structure must also exist in order to add and work with files in the vault. This can be set up manually by adding appropriate registry values and recreating the folder structure, or automatically by using the following utility.

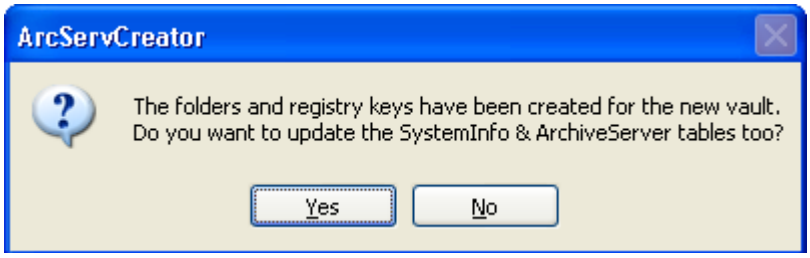
Launch the **ArcServCreator32.exe** program to create the missing archive server registry entries pertaining to the customer's vault. *Note to run the appropriate 32-bit or 64-bit version depending on operating system used.*



Enter SQL server name, ***sa*** user and password, database name (case sensitive) and archive root folder location (a subfolder named as the restored file vault will be created here to hold the vault archives).

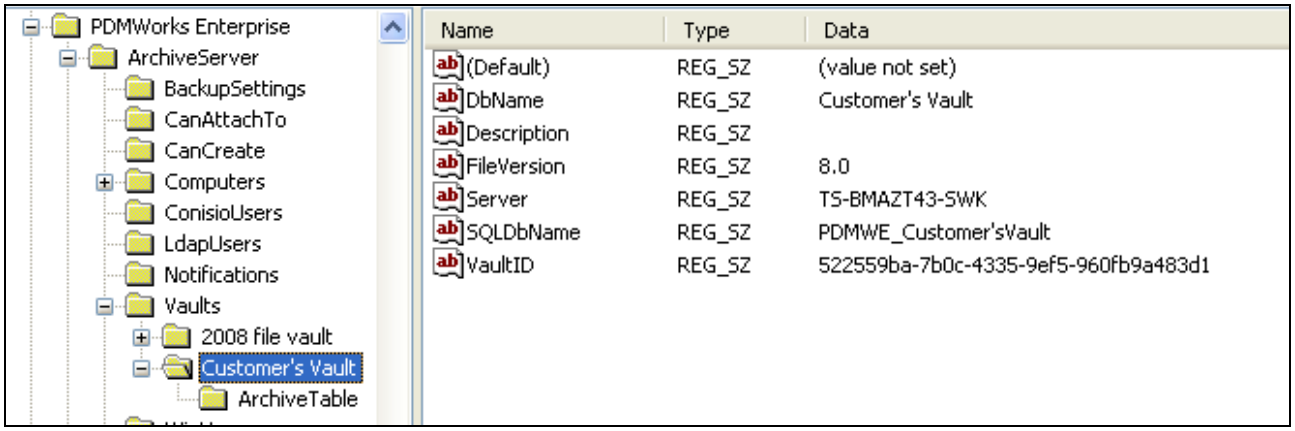


Click yes when asked to update the **SystemInfo** and **ArchiveServers** tables. This will update the tables with the current archive server name. This is required in order to attach and work with the copied vault.

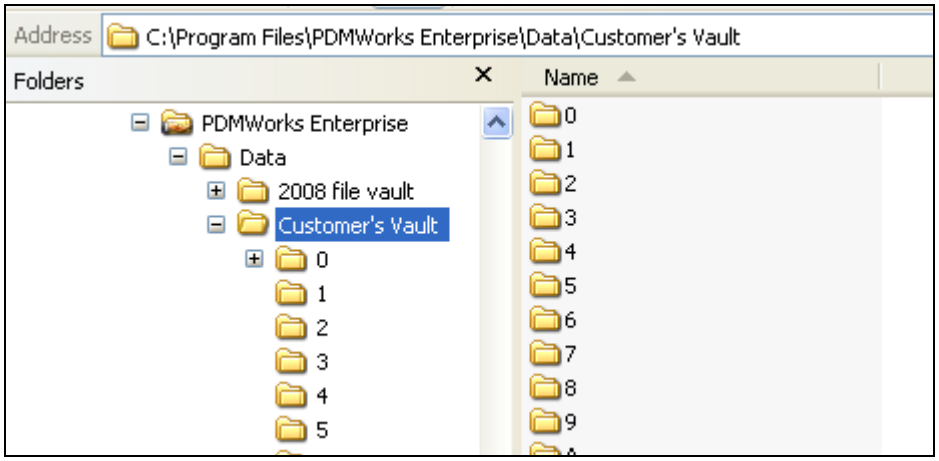


The archive server registry should now have an entry for the restored file vault under the following key:

HKEY_LOCAL_MACHINE\SOFTWARE\SolidWorks\Applications\PDMWorks Enterprise\ArchiveServer\Vaults\[vaultname]



An empty file vault archive folder structure will also be created – any new files you add to the restored file vault will be created in file archives under these folders.



8. Overwrite with original file vault archives (optionally)

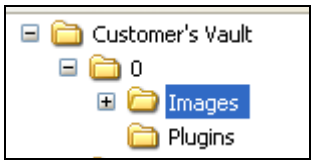
In the event you have a backup of the original file vault archives (i.e. the 16 archive subfolders named 0-F), the backup folders can overwrite the empty file vault archive folder structure that was created in the previous step.

This will allow you to create an identical copy of the original vault, including files, add-ins, card images etc. This can be useful if you wish to verify that the file vault archives upgrades to a new version for example.

It is not however required to have a copy of the original archives – you can add new files to the restored vault copy to test functionality – but since all original files will still show up when working within the restored vault you will get access errors when trying to get those files (open, preview etc). All the database information about the files will still be available.

If you do not want to restore the whole vault archive, it is recommended that you at least obtain a copy of the plugins and images folders from the original file vault archive. That will allow the file data card images and custom API add-ins in the restored vault to show up correctly.

The folders are found in the vault archive under **…\[vaultname]\0\Images** and **…\[vaultname]\0\Plugins**

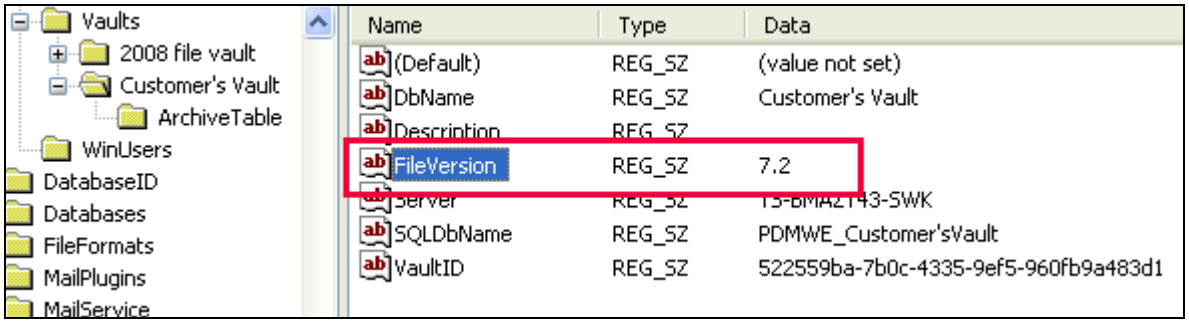


You should also ensure that the archive version in the archive server registry matches the version that the restored archives are currently in. Open the following registry key (using Regedit) on the archive server:

HKEY_LOCAL_MACHINE\SOFTWARE\SolidWorks\Applications\PDMWorks Enterprise\ArchiveServer\Vaults\[vaultname]

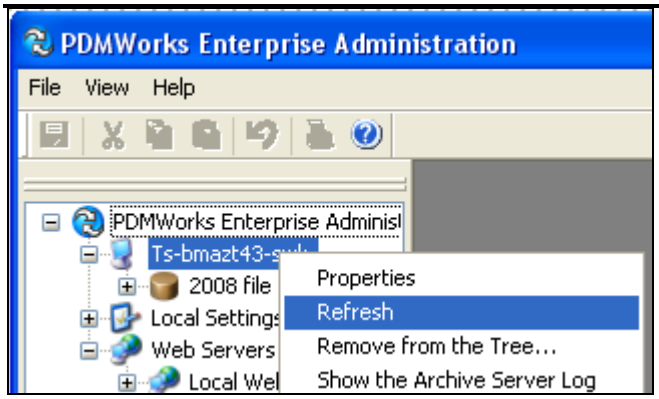
Set the **FileVersion** value to one of the following entries depending on version of the original vault archive (*note that the fileversion is not the same as the software version*).

Conisio 6.4 > PDMWE 2007	7.2
PDMWE 2008 SP0-SP3	8.0
PDMWE 2008 SP4 >	9.0

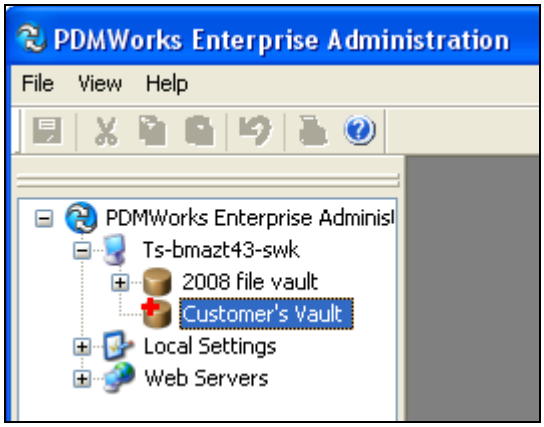


9. Attach to the restored file vault

In order to see the restored file vault in the administration tool from a client, right-click on the archive server and select **Refresh**

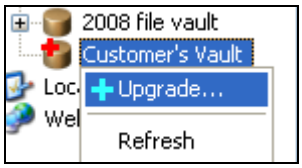


The file restored file vault copy should now show up in the tree (if it does not, verify that you have created the archive server registry entries correctly in step 7).



If the file vault is marked with a red plus icon, it indicates that the vault archive must be upgraded to match the currently installed version of PDMWE. If there is no red plus, it means that the file vault archive is of correct version and you can log into the vault and administer it.

To upgrade the archives, right-click on vault name and select **Upgrade**, log in as **admin**.



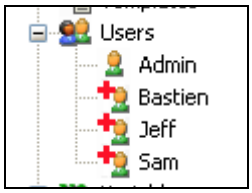
When the upgrade completes, you should be able to log into and administer the vault.

If you are having problems logging in to the vault at this point, verify the SQL login information using the archive server configuration utility.

10. Repair user logins

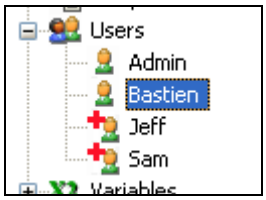
If the restored vault contained existing users that do not yet exist as login names on your archive server, the user login information (name and password) must be added to the archive server before they can be used.

Expand the **Users** node, each login that is missing on the archive server will show with a red plus sign.



To "repair" the login, open the user profile card, click **Set Password** and reset the password. This will create the missing login information in the archive server registry.

When you refresh the **Users** node, the login should now be ok to use.



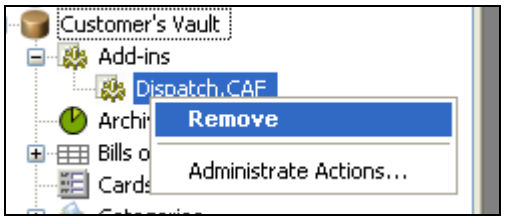
11. Remove custom add-ins and dispatch

N.B.: This step should not be done if you have already restored the **plugins** archive folder in step 8 – in which case the source add-in files already exist.

If the restored file vault database was configured to use the dispatch add-in, or any custom API add-ins, those will need to be removed from vault before you can browse in it.

In order to remove an add-in, you must first create a local file vault view. Right-click on the file vault name and select **Create local view**.

An Explorer window will appear prompting you to log in – before logging in, switch back to the administration tool and expand the **Add-ins** node. Right-click on each add-in and select **Remove**.



The restored file vault copy should now be ready to use.